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## **Imaging Enhancement of VSP Data with Optimal Aperture RTM**

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### **Abstract**

Seismic images from vertical seismic profile (VSP) data usually have higher signal-to-noise ratio and higher resolution than images produced from surface seismic data since receivers are closer to the targets. However, VSP images often contain strong migration artifacts (known as migration swings), due to the asymmetric acquisition geometry and the limited acquisition aperture. It is well known that the migration swings can be attenuated by limiting the migration aperture. In this paper, we propose a data-driven method to automatically select optimal aperture for VSP reverse time migration (RTM). Our method is based on the local coherency analysis of the migrated common image gathers (CIGs) in the source offset domain. By attenuating the long "tails" of the migrated events in the CIGs, we show that migration images can be enhanced. Numerical examples for synthetic data demonstrate the effectiveness of our proposed method for suppressing migration artifacts in VSP RTM images.

### **Introduction**

VSP data can produce higher resolution local images around the borehole compared to surface seismic data, since that receivers are placed closer to the targets and therefore can record data with higher signal-to-noise ratio and higher frequency content. VSP data are often utilized to support surface seismic processing and interpretation, providing additional information such as time/depth information, interval velocity,  $V_p/V_s$  ratio, attenuation, and anisotropy (Van Dok et al., 2021). Images from VSP data have demonstrated benefits for improving stratigraphic interpretation and fault delineation (van Gestel et al., 2019), as well as geologic CO<sub>2</sub> storage monitoring (Couëslan et al., 2013).

However, migration images from VSP data often contain strong artifacts due to the intrinsic asymmetric geometry and the narrow aperture of the VSP acquisition. Various methods have been proposed to enhance image quality of VSP data by using limited migration aperture. Lou et al. (2009) selected an appropriate migration aperture angle for Kirchhoff migration to suppress VSP migration artifacts. Ke and Shi (2018) analytically calculated the approximate range of imaging for VSP data based on the assumption of horizontal reflectors and then selected effective imaging aperture for VSP RTM based on the theoretical estimation. Hu et al. (2023) proposed a structurally adaptive aperture for RTM by limiting the range of the subsurface opening angle. Rusmanugroho and Jaya (2023) presented a Fresnel volume migration method to enhance

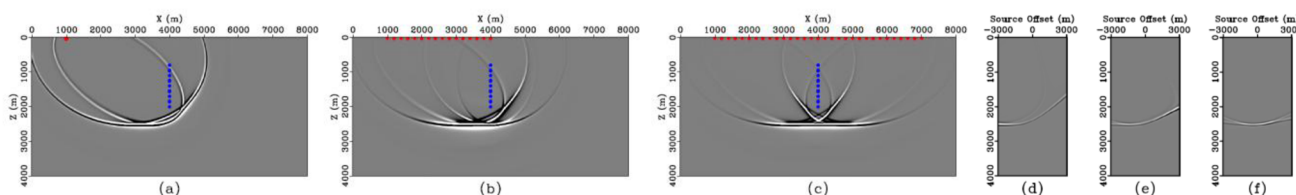
VSP imaging. Others proposed enhanced imaging workflows and algorithms to suppress migration noise in VSP images. Guo et al. (2022) developed a VSP RTM workflow in the dip angle domain to attenuate migration artifacts by distinguishing real geologic dips and dips related to artifacts. Kazei et al. (2023) compared RTM and Full Waveform Inversion (FWI) imaging for DAS VSP data and showed that FWI imaging produced VSP images with reduced artifacts.

In this paper, we propose a data-driven method for selecting optimal aperture in VSP RTM to suppress migration artifacts based on the local semblance analysis of common image gathers (CIGs). We first illustrate the migration artifacts due to the acquisition geometry of the 2D walkaway VSP data and introduce the source offset domain CIGs. Then we present our workflow of optimal aperture RTM for VSP data. Finally, numerical examples for synthetic data are shown to demonstrate the effectiveness of our proposed method.

## VSP RTM and Source Offset Domain CIGs

In this section, we use a simple two-layered model to illustrate the migration artifacts in VSP RTM due to the acquisition geometry. The model has a size of 8000 m in the horizontal direction and 4000 m in the vertical direction. The grid spacing is 10 m in both directions, and the reflector is located at the depth of 2500 m. A total of 121 shots are distributed evenly on the surface from 1000 m to 7000 m. Receivers are placed at the horizontal distance of 4000 m and at depths from 800 m to 2000 m with an interval of 10 m. In all our experiments, we apply inversion-based radon transform with sparseness constraint (Verschuur et al., 2012) to extract upgoing wavefields, and use inverse scattering imaging condition for RTM (Whitmore, and Crawley, 2012; Liang and Zhang, 2019).

Figure 1 shows the VSP RTM results using upgoing wavefields. Figure 1a is the migration result of the first shot, where the shot and receiver locations are indicated by the red and blue dots, respectively. From Figure 1a we can observe the limited illumination of the reflector and the strong migration swing noise for a single shot, both of which are caused by the limited coverage and the asymmetric geometry of the typical walkaway VSP acquisition. Figures 1b and 1c show the migration results of the first 66 shots and all the 121 shots, respectively, from which we can see that migration swings cannot be eliminated by stacking of multiples shots.



**Figure 1—VSP RTM images of the layered model. (a) Migration result of a single shot, (b) migration result of the first 66 shots, (c) migration of all the 121 shots, the source offset CIG extracted at the horizontal distance of (d) 3300 m, (e) 3600 m, and (f) 3900 m.**

Next, we extract the source offset domain CIGs. Since the horizontal location of receivers are fixed, each shot can be indexed by the source offset (the horizontal distance between source and receiver), as shown in Figure 2. For the two-layered model, three source offset CIGs are extracted and shown in Figs. 1d-1f. Unlike surface seismic data, the source offset CIGs in VSP imaging are curved and non-flat using correct migration velocity. Du et al. (2015) pointed out that the extreme point (apex) of the reflection event is located on the true reflector depth (Figure 2). While stacking over all the shots (i.e., stacking along the horizontal direction in the source offset CIGs) produces the final stacked image of reflectors at correct depth, the long "tails" of the events in Figures 1d-1f will produce migration artifacts. Inspired by previous work of selecting migration

aperture using CIGs (Giboli et al., 2013; Klovov, and Fomel, 2013; Lu et al., 2020), we propose an optimal aperture RTM method using source offset CIGs to suppress VSP migration artifacts in the next section.

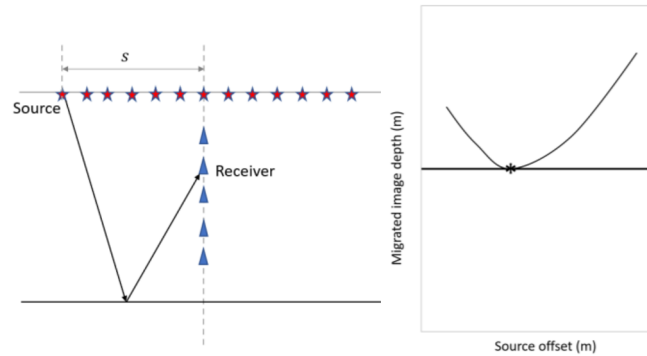


Figure 2—Left figure shows the 2D walkaway VSP acquisition geometry and right figure shows the migrated source offset domain CIG for the walkaway VSP data using correct velocity (note that the extreme point indicated by star is located on the true reflector depth).

## Optimal Aperture RTM of VSP data

In this section, we present a method to select optimal aperture for VSP RTM based on the local coherency analysis of source offset CIGs. We use synthetic data generated from the 2D SEAM Arid model as an example to demonstrate our workflow. Figure 3a shows the velocity model. For the 2D walkaway VSP acquisition, shots are located on the surface at horizontal location  $x_s$  from 2000 m to 11120 m with an interval of 40 m indicated by the red dots, and receivers (illustrated by blue dots) are placed in the well at horizontal location  $x_g = 5000$  m and at depths from 550 m to 3243.75 m with an interval of 6.25 m. Figure 3b shows the migrated image  $I_0(z, x, s)$  for all the shots before stacking, where  $z$  represents migrated depth,  $x$  represents horizontal location, and source offset  $s$  is defined as  $s = x_s - x_g$ . The final stacked image can be obtained by stacking Figure 3b along the source offset direction. For a fixed horizontal location  $x = x'$  in Figure 3b, we can extract the source offset CIG,  $I_1(z, s, x = x') = I_0(z, x, s)|_{x=x'}$ .

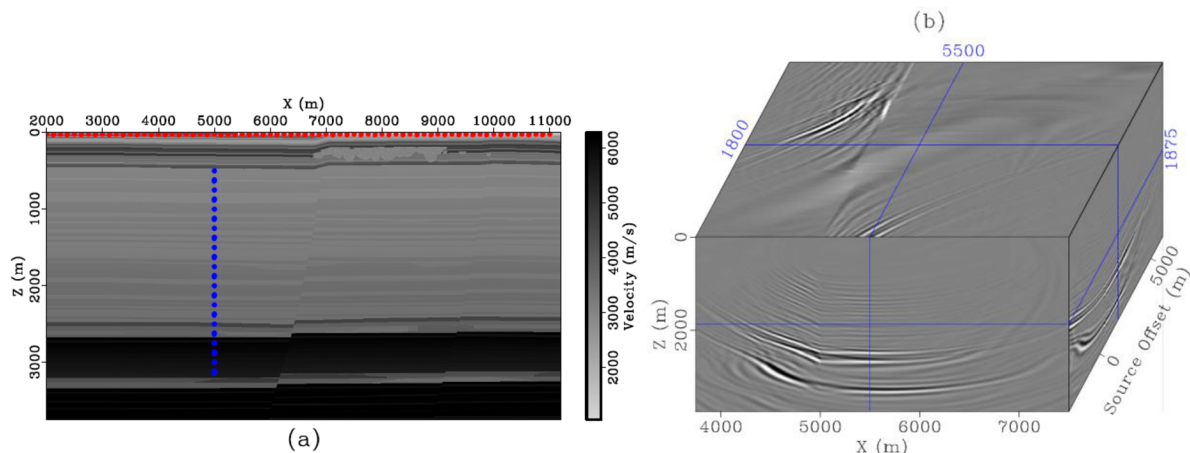


Figure 3—(a) Velocity model and (b) migrated image before stacking over shots.

By observing that diffraction events exhibit very mild dipping (quasi-linear) in the source offset domain CIGs (similar to offset-aperture domain CIGs for surface seismic data in Giboli et al. (2013)), we create a weight function based on coherence analysis to attenuate the long "tails" of reflection events while preserving reflection apex and diffractions in the CIGs. For each source offset CIG, we perform a two-step workflow to suppress migration swing noise.

The first step is designed to attenuate low frequency noise and highly dipping events (crossing events) by applying wavelet-domain filtering. This step is to prepare the CIGs for the coherence analysis in the next step. We choose the dual-tree complex wavelet transform, since it decomposes data into higher dimensions and can better facilitate the separation of signals and noise (Selesnick et al., 2005; Liang et al., 2022). The dual-tree complex wavelet transform decompose events into multi-scales and six distinct orientations ( $\pm 75^\circ$ ,  $\pm 45^\circ$ ,  $\pm 15^\circ$ ) (Figure 4). We first apply the 2D dual-tree complex wavelet domain transform to each CIG  $I_1(z, s, x)$  ( $x$  is a constant in each CIG):

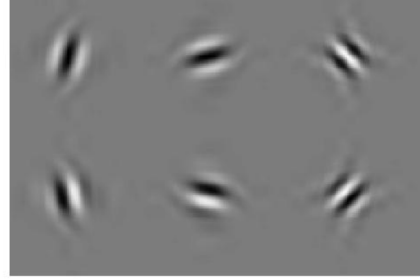


Figure 4—Wavelet basis of the 2D dual-tree wavelet transform.

$$\tilde{I}_1(z, s, x, j, o, ri) = CWT_{2D}(I_1(z, s, x)), \quad (1)$$

where  $j$  represents scale,  $o$  represents orientation,  $ri$  indicates a real or imaginary part. Then we use equation 2 to attenuate low frequency noise and highly dipping events:

$$\tilde{I}_2(z, s, x, j, o, ri) = \begin{cases} \tilde{I}_1(z, s, x, j, o, ri), & \text{for selected } j, o \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

In our experiments, a total of 5 scales are used, and we choose to mute the lowest frequency scale and events oriented in directions of  $\pm 75^\circ$ . In practical applications, the user-defined parameters can vary for different data and can be determined through trial and error. Then we apply the inverse wavelet transform to  $\tilde{I}_2(z, s, x, j, o, ri)$  to obtain the denoised CIGs  $I_2(z, s, x)$ . Figure 5a shows an example of raw CIG  $I_1(z, s, x)$  at location  $x = 4525$  m, which only contains reflection events. Figure 5b shows the CIG after applying wavelet-domain denoising.

The second step is to measure the local coherence of the CIGs  $I_2(z, s, x)$  (Klokov and Fomel, 2013; Liu and Zhang, 2019) and get the semblance function  $C(z, s, x)$  for each fixed  $x$  as shown in equation 3:

$$C(z, s, x) = \frac{\sum_{z'=z-\Delta h_1}^{z+\Delta h_1} \left( \sum_{s'=s-\Delta h_2}^{s+\Delta h_2} I_2(z', s', x) \right)^2}{N \sum_{z'=z-\Delta h_1}^{z+\Delta h_1} \sum_{s'=s-\Delta h_2}^{s+\Delta h_2} (I_2(z', s', x))^2}, \quad (3)$$

where  $\Delta h_1$  and  $\Delta h_2$  are the half window size in the  $z$  and  $s$  direction, respectively, and  $N$  is the total number of traces in each CIG.

High semblance values indicate signals with better horizontal coherence, which correspond to apex of reflections and diffractions events in the CIGs. Next we apply the following thresholding in equation 4 to the semblance function and obtain the weight function for CIGs:

$$w(z, s, x) = \begin{cases} 1, & \text{for } C(z, s, x) \geq c_2 \\ \frac{C(z, s, x) - c_1}{c_2 - c_1}, & \text{for } c_1 \leq C(z, s, x) < c_2 \\ 0, & \text{for } C(z, s, x) < c_1 \end{cases} \quad (4)$$

where  $c_1$  and  $c_2$  define two thresholds. Applying the weight function  $w(z, s, x)$  to the filtered CIGs  $I_2(z, s, x)$  as shown in equation 5 corresponds to automatically selecting migration aperture while preserving areas near reflection apex and diffractions.

$$I_3(z, s, x) = w(z, s, x) \cdot I_2(z, s, x). \quad (5)$$

Figure 5c shows the final weight function after applying thresholding in equation 4 to the semblance function. The final CIG  $I_3(z, s, x)$  after applying weighting is shown in Figure 5d. By comparing results in Figure 5a and Figure 5d, we observe that the long "tails" of reflection events are attenuated. Figure 6 illustrates the results when the same procedure is applied to a second raw CIG including both reflection and diffraction events at location  $x = 6400$  m. Results from Figure 6 demonstrates that both reflection apex and diffraction events are preserved by the proposed method.

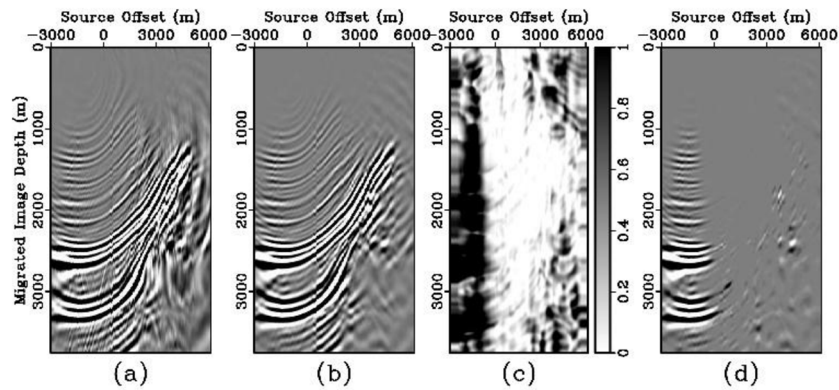


Figure 5—(a) Raw source offset domain CIG at location  $x = 4525$  m only including reflection events, (b) CIG after applying wavelet domain filtering, (c) the weight function, and (d) the CIG after applying the weight function.

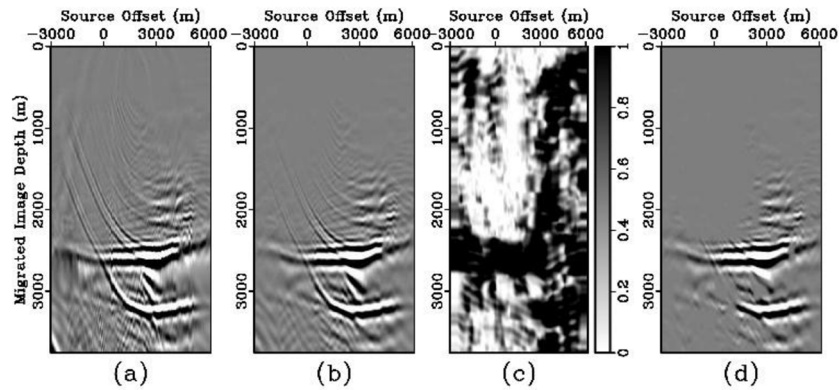


Figure 6—(a) Raw source offset domain CIG at location  $x = 6400$  m including both reflection and diffraction events, (b) the weight function, and (c) the CIG after applying weight function.

The final denoised image is obtained by stacking the denoised CIGs  $I_3(z, s, x)$  along the source offset direction:

$$I_{\text{Final}}(z, x) = \int I_3(z, s, x) ds. \quad (6)$$

## Results

In this section, the workflow described in the previous section is applied to all the source offset CIGs from the 2D SEAM Arid synthetic data. We examine six CIGs from different locations, and show the comparison of the raw CIGs, wavelet-domain filtered CIGs, and weighted CIGs using equation 5 in Figure 7. Stacking



all the CIGs along the source offset direction generates the stacked images. [Figure 8a](#) shows the stacked image of raw CIGs. In comparison, the stacked image of CIGs after wavelet-domain filtering and after weighting using [equation 5](#) are illustrated in [Figure 8b](#) and [Figure 8c](#), respectively. The final denoised image in [Figure 8c](#) exhibits reduced artifacts and enhanced image quality compared to raw stacked image in [Figure 8a](#), especially near the well location.

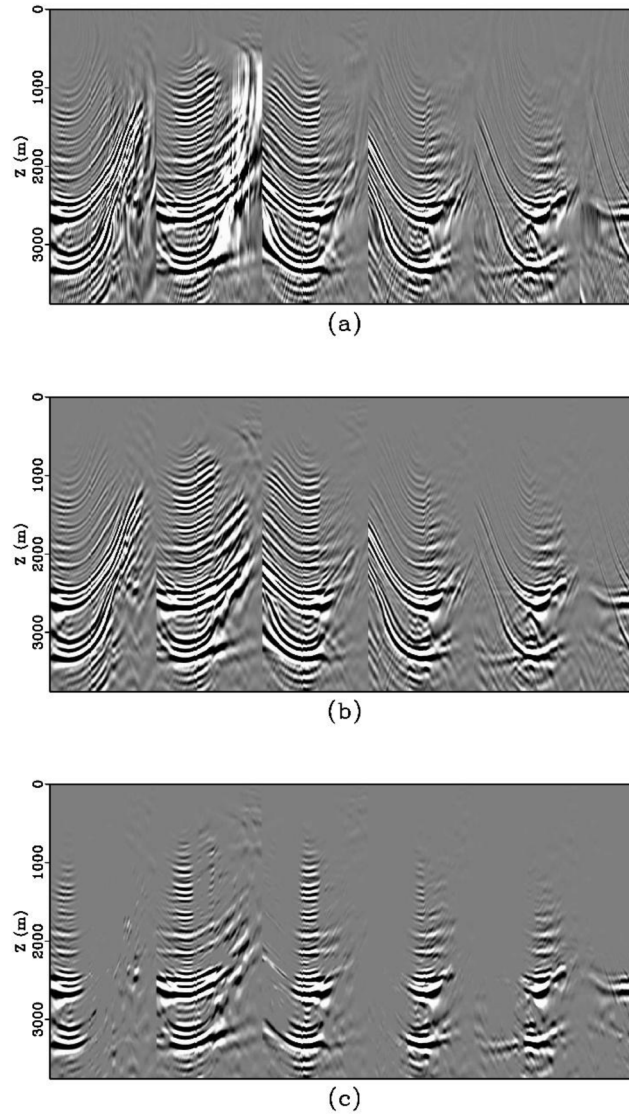


Figure 7—(a) Raw source offset CIGs extracted from [Fig. 3b](#) at six horizontal locations (4500 m, 4875 m, 5250 m, 5625 m, 6000 m, 6375 m), (b) the corresponding CIGs after wavelet-domain filtering, and (c) CIGs after applying the weight function in [equation 4](#) to [Fig. 7b](#).

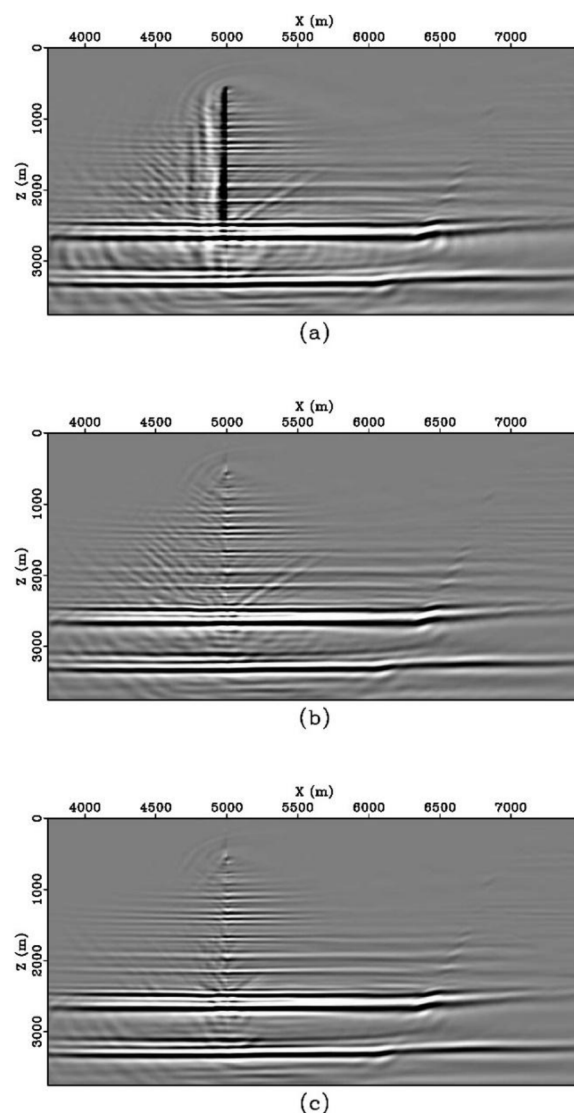


Figure 8—Stacked image of (a) raw CIGs, (b) CIGs after wavelet-domain filtering, and (c) CIGs after applying the weight function.

Next, we apply the proposed method to the modified Sigsbee2B model, with a salt body embedded within a simple sedimentary section. The model has a width of 16000 m and a depth of 6000 m, with a grid spacing of 10 m in both directions. A portion of the velocity model is shown in Figure 9a. We generate synthetic data using 121 shots located on the surface from 3000 m to 9000 m. Receivers are placed at the horizontal distance of 6000 m and at depths from 500 m to 3000 m with an interval of 10 m. We use the velocity model in Figure 9b for migration. Figures 9c and 9d show the raw stacked image, and the denoised stacked image using the proposed method, respectively. Comparison of the raw stacked image and the final denoised image from Figure 9 demonstrates the effectiveness of our proposed method for suppressing VSP RTM migration artifacts.

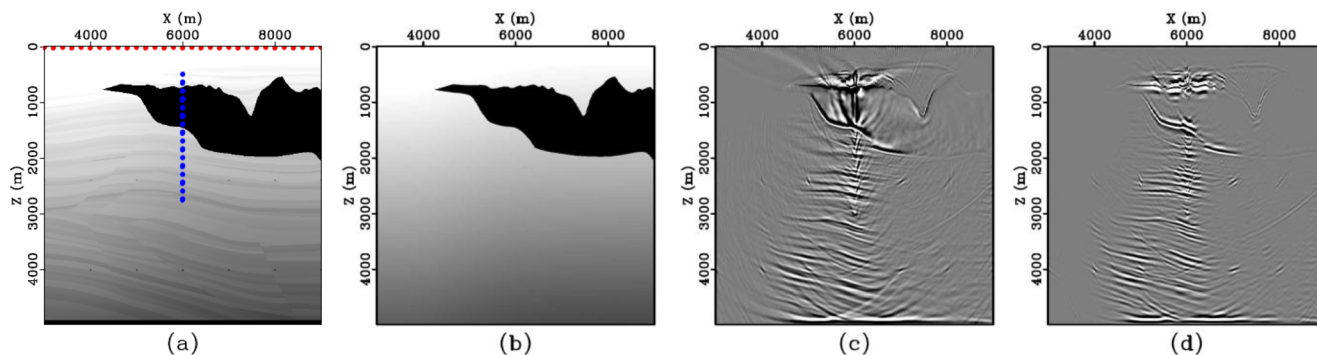


Figure 9—(a) True velocity model, (b) migration velocity model, (c) raw stacked image, and (d) denoised stacked image after applying the proposed method.

Finally, the workflow is tested on the Marmousi model with highly-dipping reflectors in Figure 10. The acquisition system includes 301 shots located on the surface from 1000 m to 7000 m and 101 receivers at the horizontal distance of 4000 m and at depths from 500 m to 1500 m. The denoised image in Figure 10c exhibits improved delineation of reflectors after artifacts attenuation compared to the raw image in Figure 10b.

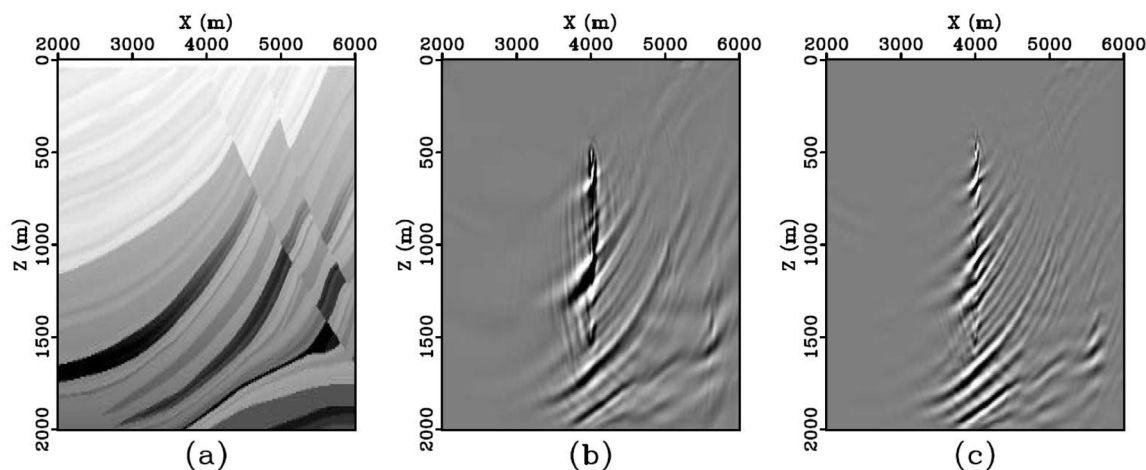


Figure 10—(a) A portion of Marmousi velocity model, (b) raw stacked image, and (c) denoised stacked image.

## Conclusion

We propose a new method to select an optimal aperture for VSP RTM to suppress migration artifacts due to its asymmetric geometry and limited coverage of the acquisition. Our method utilizes migrated CIGs in the source offset domain and apply a two-step workflow to each CIG. The first step applies wavelet-domain denoising to the CIGs, and the second step attenuates the long "tails" of the events in the CIGs based on the local coherence analysis. The final denoised image is obtained by stacking all the CIGs. Our proposed workflow requires no prior information of the subsurface model and automatically selects optimal aperture for VSP RTM image. Numerical tests on three synthetic models demonstrate that our proposed method can effectively suppress migration artifacts and enhance image quality for VSP RTM images.

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